

GEILO-OLDEBRATEN, Norway

WMO: 013590

Lat: 60.530N

Lon: 8.195E

Elev: 773

StdP: 92.38

Time Zone: 1.00 (EUC)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -24.0	(c) -21.1	(d) -26.8	(e) 0.4	(f) -23.6	(g) -24.0	(h) 0.5	(i) -20.9	(j) 11.3	(k) 0.1	(l) 9.8	(m) 0.6	(n) 0.6	(o) 220	(p) 0.711

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.0	22.8	13.4	20.6	12.9	18.5	11.8	14.8	20.2	13.8	18.5	12.8	17.1	2.5	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.8	10.1	16.0	11.7	9.4	15.1	10.7	8.8	14.1	43.8	20.3	41.0	18.4	38.4	17.2	18.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 9.0	(b) 7.5	(c) 6.4	(e) -27.3	(f) 25.2	(g) 3.6	(h) 2.1	(i) -29.8	(j) 26.8	(k) -31.9	(l) 28.0	(m) -33.9	(n) 29.2	(o) -36.5	(p) 30.8		
			DB	WB	WB	WB	WB	WB	WB	WB	WB	WB	WB	WB		
			(e) -27.3	(f) 15.9	(g) 3.6	(h) 1.2	(i) -30.0	(j) 16.7	(k) -32.1	(l) 17.4	(m) -34.1	(n) 18.1	(o) -36.7	(p) 19.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.0	-7.4	-6.4	-3.3	0.6	5.6	9.9	12.4	10.9	7.5	2.3	-2.7	-5.6
	DBStd	8.20	6.47	6.14	5.16	3.71	3.75	2.80	2.96	2.34	2.83	3.57	5.04	6.63
	HDD10.0	3088	539	458	412	283	146	35	9	17	83	241	382	484
	HDD18.3	5952	797	692	670	533	394	254	183	231	324	498	632	743
	CDD10.0	178	0	0	0	0	9	31	84	44	9	1	0	0
	CDD18.3	1	0	0	0	0	0	0	1	0	0	0	0	0
	CDH23.3	34	0	0	0	0	1	3	31	0	0	0	0	0
	CDH26.7	2	0	0	0	0	0	0	2	0	0	0	0	0
(14) Wind	WSAvg	2.5	2.5	2.5	3.1	2.8	2.9	2.5	2.2	2.0	2.6	2.5	2.3	2.7
(15) Precipitation	PrecAvg	776	72	47	47	42	48	67	84	87	70	76	73	63
	PrecMax	1070	156	102	134	96	124	175	190	158	136	168	154	135
	PrecMin	534	11	6	7	13	4	10	17	28	33	24	18	10
	PrecStd	109	32	24	24	23	28	37	41	37	24	35	31	26
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.6	6.7	10.0	13.1	21.8	23.5	26.4	22.2	19.1	14.5	8.2	6.1
		MCWB	3.1	3.4	5.3	6.8	11.9	12.6	15.2	13.7	12.0	10.4	5.4	4.3
	2%	DB	4.1	4.7	6.5	10.2	18.0	21.3	23.9	19.7	16.7	11.2	6.3	5.0
		MCWB	2.0	2.2	3.1	5.1	10.6	12.1	14.4	12.8	10.9	7.2	4.2	3.2
	5%	DB	2.7	3.4	4.8	8.1	15.3	18.8	21.8	17.9	14.6	9.3	4.8	3.9
		MCWB	0.8	1.2	2.1	4.1	9.2	10.6	13.8	12.2	10.0	6.3	2.9	2.0
	10%	DB	0.9	1.8	3.4	6.5	12.6	16.3	19.3	16.0	12.5	7.7	3.6	2.6
		MCWB	-0.8	-0.3	0.8	3.2	7.5	9.7	12.7	11.5	9.1	5.2	1.8	0.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.4	3.8	5.3	7.4	13.3	13.8	16.7	15.1	13.5	11.0	5.9	4.5
		MCDB	5.3	6.1	9.0	11.7	18.9	19.1	22.8	19.2	17.1	14.5	7.9	5.9
	2%	WB	2.2	2.5	3.4	5.7	11.2	12.8	15.4	14.0	11.7	8.1	4.2	3.2
		MCDB	3.9	4.5	6.0	9.4	16.7	18.9	21.3	17.7	15.0	10.5	5.8	4.8
	5%	WB	0.8	1.3	2.4	4.5	9.7	11.8	14.5	13.0	10.6	6.7	3.1	2.2
		MCDB	2.6	3.3	4.7	7.8	14.3	17.2	20.2	16.7	13.5	8.8	4.6	3.8
	10%	WB	-0.7	-0.3	0.9	3.4	8.2	10.7	13.5	12.0	9.5	5.3	1.9	0.8
		MCDB	0.8	1.7	3.2	6.1	12.4	15.2	18.3	15.3	12.1	7.4	3.5	2.4
(35) Mean Daily Temperature Range	MDBR		7.4	8.1	8.3	8.4	8.9	9.9	10.0	9.3	8.3	7.0	6.3	6.9
	5% DB	MCDBR	7.2	7.0	8.0	11.3	13.5	15.3	15.5	12.9	12.4	10.3	7.0	6.3
		MCWBR	6.2	5.6	5.7	7.1	7.4	7.4	6.9	6.7	7.4	6.9	5.6	5.6
	5% WB	MCDBR	7.2	6.7	7.1	10.1	12.4	12.0	13.1	10.8	10.3	8.2	6.5	6.4
		MCWBR	6.3	5.6	5.4	6.4	7.1	6.5	6.2	6.2	6.6	6.0	5.3	5.7
(40) Clear-Sky Solar Irradiance	taub		0.210	0.250	0.283	0.317	0.324	0.321	0.335	0.327	0.297	0.279	0.222	0.180
	taud		2.156	2.269	2.242	2.240	2.363	2.471	2.455	2.464	2.539	2.458	2.201	2.023
	Ebn at Noon		588	768	847	879	901	911	886	866	842	732	557	454
	Edh at Noon		37	63	91	112	107	98	98	89	69	53	35	26
(44) All-Sky Solar Radiation	RadAvg		0.30	0.92	2.35	4.06	4.79	5.05	4.62	3.63	2.36	1.10	0.38	0.18
	RadStd		0.06	0.22	0.26	0.59	0.62	0.52	0.58	0.32	0.30	0.19	0.08	0.02

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (55 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page